

A Hierarchical Architecture for Humanoid Hand Reaching with Guided Diffusion Planning

Yumeng Liu¹, Yijun Fan¹, Jiyu Yu¹, Hui Chai³, Tongye Xu², Rong Xiong¹, and Yue Wang¹

Abstract—Task-oriented humanoid reaching requires both human-like whole-body motion and robust closed-loop execution. Existing approaches are limited by two major challenges: the difficulty of obtaining scalable robot-side task data, and the mismatch between high-level motion planning and low-level whole-body control. To alleviate these issues, we propose a hierarchical framework that integrates task-oriented data generation, guided diffusion planning, and whole-body tracking control. A task-oriented robot-side dataset is constructed from a physically grounded latent motion prior and goal relabeling. A guided diffusion planner is then trained to predict future sparse keypoint trajectories, with additional robustness provided by noise augmentation and Dagger-style out-of-distribution post-training. Combined with a teacher-student whole-body controller, the resulting system enables stable humanoid reaching in closed loop. Experiments on a Unitree H1 humanoid in Isaac Gym demonstrate the effectiveness of the proposed framework under both low-dimensional and image observations, achieving an 88.89% success rate on image-conditioned reaching tasks.

I. INTRODUCTION

Task-oriented humanoid reaching requires both natural whole-body motion and robust closed-loop execution. Unlike fixed motion playback or hand-designed controllers[1], a humanoid reaching policy must autonomously coordinate perception, planning, and control under embodiment constraints. This is particularly challenging for whole-body humanoid systems, where even a simple target-reaching task may require coordinated torso adjustment, stepping, balance maintenance, and arm motion[2]. As a result, generating motions that are simultaneously task-directed, physically feasible, and human-like remains a fundamental challenge for humanoid visuomotor control.

A key bottleneck lies in data. Direct robot-side demonstration collection such as teleoperation is expensive and difficult to scale[3], especially when synchronized egocentric observations and action trajectories are all required. Although large-scale human motion datasets provide abundant motion priors, retargeted motions are often kinematically plausible but not dynamically feasible for humanoid robots due to embodiment mismatch, actuation limits, and dynamic constraints. This creates a gap between abundant human motion resources and robot-usable task supervision. Bridging this gap requires a learning framework that can convert

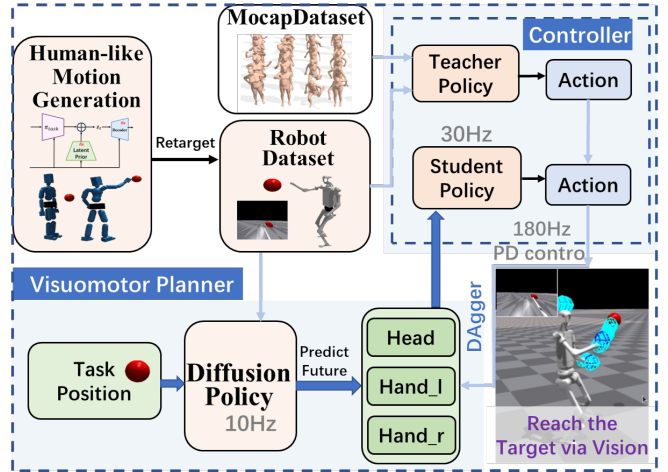


Fig. 1: Overview of the proposed framework. Task-oriented reaching motions are first generated from a human-like motion prior and retargeted to the humanoid embodiment to construct a robot-side dataset. A diffusion-based visuomotor planner then predicts future sparse keypoint trajectories from task observations, while a hierarchical teacher-student controller converts these sparse targets into executable whole-body actions for stable humanoid reaching.

human motion priors into task-oriented humanoid data that preserve both motion while preserving motion naturalness and physical plausibility, without relying on costly robot-side data collection for each downstream task.

Beyond the data issue, another challenge is policy learning and execution. Diffusion-based planners are attractive for humanoid tasks because they can model multimodal future motion distributions and generate temporally coherent trajectories over a planning horizon. However, directly applying diffusion policies to humanoid systems remains nontrivial. First, rollout errors may accumulate during closed-loop execution, causing the planner to encounter out-of-distribution (OOD) observations that deviate from the offline training manifold. [4], [5] Second, the generated sparse motion plan must ultimately be executed by a physically grounded controller, and mismatch between planning outputs and low-level whole-body execution may lead to jitter, discontinuity, or task failure. Therefore, improving OOD robustness during rollout and integrating planning with stable whole-body control are both critical.

To address these challenges, we propose a hierarchical framework for humanoid reaching that integrates scalable robot-side data generation, sparse diffusion planning, and whole-body tracking control. We first construct task-oriented robot data by retargeting motions synthesized from a pre-trained human motion prior and relabeling goals using realized motion outcomes. Based on this dataset, a guided

¹Yumeng Liu, Yijun Fan, Jiyu Yu, Rong Xiong and Yue Wang are with the State Key Laboratory of Industrial Control and Technology, Zhejiang University, Hangzhou 310027, China. Yue Wang is the corresponding author (e-mail: wangyue@iipc.zju.edu.cn).

²Tongye Xu is with the College of Information Science and Engineering, Northeastern University, China.

³Hui Chai is with the School of Control Science and Engineering, Shandong University, China.

diffusion planner predicts future sparse head-and-hand trajectories from recent observations, with improved rollout robustness from noise augmentation and DAgger-style refinement. These sparse plans are then executed by a teacher-student whole-body controller for stable closed-loop reaching. Experiments in Isaac Gym on a Unitree H1 humanoid validate the effectiveness of the proposed framework under both low-dimensional and image observations.

The main contributions of this work are as follows:

- We propose a scalable task-oriented data generation pipeline that converts a human motion prior into robot-side reaching data through retargeting and goal relabeling.
- We develop a guided diffusion planner for sparse keypoint prediction, and improve its robustness with noise augmentation and DAgger-style post-training.
- We integrate the planner with a whole-body controller within a hierarchical framework and validate the complete system in simulation.

II. RELATED WORK

A. Task-oriented Data Generation for Humanoid Learning

Learning visuomotor humanoid policies requires large amounts of task-relevant demonstrations. A straightforward solution is teleoperation or direct robot data collection, which provides paired observations and actions for policy learning. However, such data are expensive to collect and difficult to scale, especially for humanoids, where whole-body motion, balance, and perception must be captured jointly.

An alternative direction is to leverage large-scale human motion datasets, such as AMASS [6], and convert them into robot-usable supervision. Prior work has shown that motion priors learned from human motion can provide reusable and natural movement representations for downstream control. In particular, physics-based trackers such as PHC and motion-prior distillation methods such as PULSE are relevant because they ground motion generation in physically simulated tracking and produce dynamically meaningful latent skill spaces [7], [8]. Compared with purely kinematic motion generation[9], these physics-grounded approaches are better suited for transfer to embodied humanoid control.

Recent work has explored constructing paired state-action datasets in simulation by tracking motion sequences with physics-based policies for imitation learning [10]. However, these methods focus on virtual characters rather than real humanoid embodiments. This indicates that simulation-based motion generation and retargeting can bridge human motion data and humanoid policy learning. Different from generic dataset curation, our work follows a task-oriented pipeline: we generate task-consistent virtual humanoid motions, retarget them to the humanoid robot, and use the resulting robot trajectories for downstream policy learning.

B. Diffusion-Based Planning and Hierarchical Execution

Diffusion models have become a promising paradigm for robot policy learning because they can represent multimodal

action distributions and generate temporally coherent trajectories over long horizons [11]. This makes them attractive for humanoid motion planning, where behaviors are highly multimodal and constrained by both task objectives and embodiment.

However, diffusion-based humanoid planning remains difficult due to the high-dimensional whole-body action space, strong dynamic coupling, and strict execution constraints. Hierarchical planner-controller frameworks provide a practical solution by combining low-frequency planning with high-frequency execution. Yet, when the two modules are separated, mismatch between generated trajectories and physically executable motions can cause unstable behaviors, while rollout errors may accumulate and push the planner into out-of-distribution(OOD) states [12], [10]. In our work, we retain this hierarchical structure and further improve it through guided diffusion planning and OOD post-training, so that high-level plans are better matched to downstream whole-body control.

III. METHODOLOGY

A. Framework Overview

As illustrated in Fig. 1, the framework consists of three components: task-oriented dataset generation, sparse diffusion planning, and whole-body tracking control. The planner predicts future head-and-hand trajectories from recent observations, and the controller tracks them with executable joint commands. We instantiate this framework on a humanoid ball-reaching task, where the robot must locate the target and reach it with the right hand while maintaining stable whole-body motion.

B. Scalable Reaching Dataset from Motion Prior and Goal Relabeling

To avoid expensive robot-side teleoperation, we construct a scalable dataset for a right-hand ball-reaching task from a physically grounded motion prior learned by PULSE, and then transfer the generated motions to the target humanoid embodiment through retargeting.

Our starting point is a latent motion prior distilled from a universal motion tracker trained via physics-based tracking of large-scale human motion data. Because this prior is grounded in physically consistent whole-body tracking, it captures coordinated, human-like, and dynamically plausible behaviors in a compact latent skill space. We first instantiate the reaching task on a simplified virtual humanoid embodiment, where a target ball is placed in the workspace and task-conditioned motions are synthesized from the prior. The high-level policy is trained with PPO[13] to drive the right hand toward the target while preserving stable and human-like whole-body behavior. A terminal reward further distinguishes successful reaching from failure cases. The resulting virtual motions then serve as the source trajectories for retargeting to the humanoid robot.

The synthesized trajectories are subsequently retargeted to the humanoid robot through an optimization-based correspondence between human-body keypoints and robot

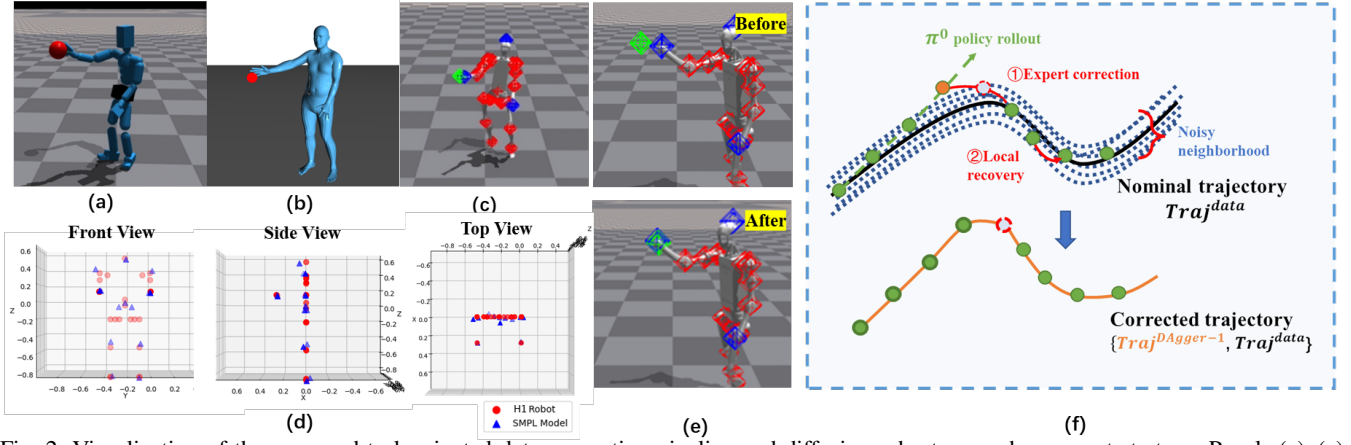


Fig. 2: Visualization of the proposed task-oriented data generation pipeline and diffusion robustness enhancement strategy. Panels (a)–(c) show synthesized human-like reaching motions and their retargeting to the humanoid embodiment. Panel (d) compares the retargeted trajectories before and after refinement from the front, side, and top views. Panel (e) illustrates HER-based goal relabeling. Panel (f) shows the proposed diffusion robustness enhancement strategy with noise augmentation and expert-guided DAGger correction.

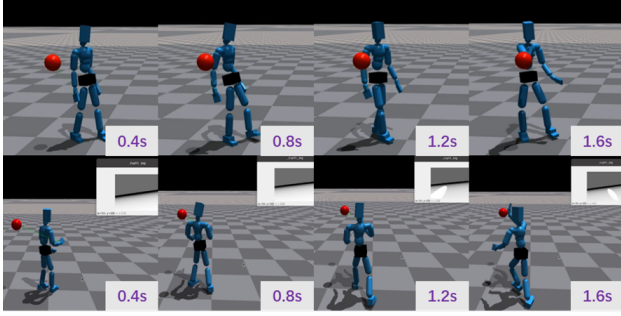


Fig. 3: Example of task-conditioned reaching motions generated on the simplified virtual humanoid. The high-level policy drives the right hand toward the target ball while maintaining stable and human-like whole-body behavior.

kinematics[14], as illustrated in Fig. 2(a)–(d). To improve motion quality, we apply temporal filtering before and after retargeting to suppress jitter and enhance smoothness. The retargeted trajectories are then replayed in simulation to collect synchronized robot observations, including proprioceptive states, task observations, and sparse action labels defined by the positions and velocities of the head and hands. Following this procedure, we obtain a scalable robot-side visuomotor dataset without requiring manual teleoperation.

A practical issue is that small mismatches are inevitably introduced during motion generation and embodiment transfer. As a result, the final realized hand position in robot space may not exactly coincide with the nominal ball position, which would otherwise create label inconsistency in policy training. To address this, we apply HER-style goal relabeling [15], replacing the original ball position with the final stabilized right-hand position as the goal label for each replayed reaching trajectory. As shown in Fig. 2(e), we improves label consistency by aligning the supervision target with the realized motion outcome. This robot-side dataset directly provides the observation-action pairs for training the diffusion planner.

C. Guided Diffusion Planning and OOD Post-Training

We formulate the high-level planner as a conditional diffusion policy that predicts future sparse keypoint trajectories

from recent observations. At each planning step, it takes a short observation history as input and predicts a future sparse trajectory over the planning horizon. In our implementation, the motion targets are the positions and velocities of the head and both hands.

Let $o_{t-H_o+1:t} = \{o_{t-H_o+1}, \dots, o_t\}$ and $X_{t+1:t+H_p} = \{x_{t+1}, \dots, x_{t+H_p}\}$ denote the observation history and predicted trajectory, where H_o and H_p are the observation and prediction horizons, respectively.

For the vision-based setting, each observation consists of a proprioceptive state and an egocentric image feature:

$$o_t = \left[o_t^{\text{prop}}, o_t^{\text{img}} \right], \quad (1)$$

where o_t^{img} denotes the visual feature extracted from the egocentric RGB image, and o_t^{prop} contains the robot proprioceptive information, including root state, local body state, joint state, and the previous action.

Each predicted token is defined as

$$x_t = \left[p_t^{\text{kp}}, v_t^{\text{kp}} \right], \quad (2)$$

where p_t^{kp} and v_t^{kp} denote the sparse keypoint positions and velocities of the head, left hand and right hand.

a) *Diffusion formulation and training:* Following Diffusion Policy, we model future sparse-keypoint generation as a conditional action diffusion process, where a sequence of future motion tokens is generated by iterative denoising under observation conditioning. Instead of directly regressing a single future action, the planner predicts a short horizon of sparse keypoint trajectories, which is better suited for capturing multimodal future behaviors and temporal consistency.

Given a clean trajectory X_0 , a noisy sample at diffusion step k is constructed as $X_k = \sqrt{\bar{\alpha}_k} X_0 + \sqrt{1 - \bar{\alpha}_k} \epsilon$, $\epsilon \sim \mathcal{N}(0, \mathbf{I})$, where $\bar{\alpha}_k = \prod_{j=1}^k \alpha_j$. We use a Transformer-based temporal denoising network $\epsilon_\theta(X_k, k, o_{t-H_o+1:t})$ to predict the injected noise conditioned on the recent observation history. Compared with direct regression, this formulation better models the multimodal distribution of future motions, while the Transformer architecture improves temporal modeling over the predicted horizon.

The denoising network is trained to predict the injected noise under observation conditioning by minimizing

$$\mathbb{E}_{X_0, \epsilon, k} \left[\|\epsilon - \epsilon_\theta(X_k, k, o_{t-H_o+1:t})\|_2^2 \right]. \quad (3)$$

At inference time, the planner starts from Gaussian noise and progressively denoises the trajectory to obtain a future sparse motion plan, which is then executed in a receding-horizon manner for closed-loop control.

b) Loss-guided denoising: To improve task alignment during inference, we introduce gradient-based guidance [16] into the reverse diffusion process. At each denoising step, the prediction is refined by a task-oriented loss

$$\mathcal{L}_{\text{guide}} = \lambda_{\text{sm}} \mathcal{L}_{\text{sm}} + \lambda_{\text{init}} \mathcal{L}_{\text{init}}, \quad (4)$$

where $\mathcal{L}_{\text{sm}}$ enforces trajectory smoothness and $\mathcal{L}_{\text{init}}$ constrains the first predicted hand state to match the current observed hand state.

The smoothness term regularizes local temporal transitions of the generated trajectory by penalizing abrupt changes between consecutive predicted tokens. Let $\hat{x}_0^i$ denote the predicted clean token at future step i . We define

$$L_{\text{sm}} = \frac{1}{N-1} \sum_{i=1}^{N-1} \|\hat{x}_0^{i+1} - \hat{x}_0^i\|_2^2, \quad (5)$$

which encourages temporally smooth sparse motion plans and suppresses jitter during replanning and execution.

and the initial-state consistency term is

$$\mathcal{L}_{\text{init}} = \|\hat{p}_{\text{hand}}^1 - p_{\text{hand}}^{\text{obs}}\|_2^2. \quad (6)$$

The guided prediction is then updated as

$$\hat{X}_0^{\text{guide}} = \hat{X}_0 - \lambda_g \nabla_{X_k} \mathcal{L}_{\text{guide}}(\hat{X}_0), \quad (7)$$

where λ_g is the guidance scale. This guidance encourages smoother trajectories and reduces discontinuities at replanning boundaries.

c) Diffusion robustness enhancement: While the guidance loss improves task alignment at inference time, we further enhance rollout robustness through two complementary strategies, as illustrated in Fig. 2(f). In closed-loop execution, accumulated tracking error, retargeting mismatch, and visual ambiguity may drive the planner into off-manifold states and degrade planning quality.

First, Gaussian noise augmentation is applied during training to both visual and proprioceptive inputs:

$$\tilde{x}_{\text{img}} = x_{\text{img}} + \epsilon_{\text{img}}, \quad \epsilon_{\text{img}} \sim \mathcal{N}(0, (0.2 \sigma_{\text{base}})^2), \quad (8)$$

$$\tilde{x}_{\text{state}} = x_{\text{state}} + \epsilon_{\text{state}}, \quad \epsilon_{\text{state}} \sim \mathcal{N}(0, \sigma_{\text{base}}^2), \quad (9)$$

which improves local robustness to mild perturbations.

Second, we perform DAGger-style [17] closed-loop refinement in simulation. Let a_t^{policy} denote the planner output and a_t^{expert} denote a heuristic recovery action. Given the rollout deviation e_t , the corrected action is defined as

$$\tilde{a}_t = \begin{cases} a_t^{\text{policy}}, & e_t \leq \delta, \\ \frac{1}{2} (a_t^{\text{policy}} + a_t^{\text{expert}}), & e_t > \delta, \end{cases} \quad (10)$$

where δ is the deviation threshold. Intuitively, the expert correction is activated only when the rollout leaves the nominal region, guiding the trajectory back toward the reference manifold. The corrected samples are aggregated with the original dataset for iterative refinement, which improves recovery from off-manifold states and reduces performance degradation under unseen task configurations.

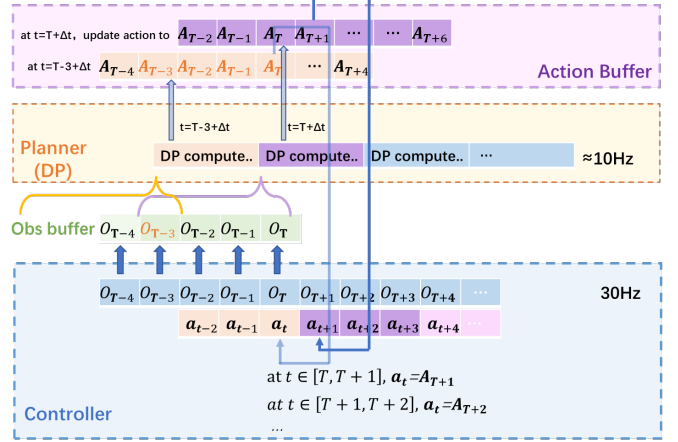


Fig. 4: Asynchronous receding-horizon execution. The diffusion planner updates future sparse targets at 10 Hz, while the low-level controller tracks the nearest valid target at 30 Hz. Only a short prefix is executed before replanning.

d) Receding-horizon planning: As illustrated in Fig. 4, we combine guided diffusion sampling with an asynchronous receding-horizon execution scheme. At each planning step, the planner takes the most recent $H_o = 4$ observation frames as input, predicts a longer horizon of future sparse trajectories, and writes the resulting waypoints together with their execution timestamps into an action buffer. Rather than executing the entire predicted horizon, the system only tracks a short prefix before replanning. This design balances long-horizon consistency and short-horizon responsiveness: a longer prediction horizon encourages coherent target-directed motion, while short-prefix execution enables rapid correction under state deviation, perception noise, and tracking error.

The planner runs at a lower frequency than the tracking controller. Therefore, after each planner update, multiple future waypoints are buffered for execution, while the controller runs several times within the same planning interval. At each control step, the controller queries the buffer and selects the nearest valid waypoint whose timestamp is greater than the current time. In this way, low-rate diffusion planning and high-rate whole-body tracking are integrated into a consistent closed-loop control process.

D. Teacher-Student Whole-Body Tracking

The planner produces sparse head-and-hand trajectories at a relatively low frequency. To execute them on a real humanoid, we use a teacher-student whole-body tracking controller. The teacher is trained with privileged simulation information, including full kinematic and dynamic states, and learns robust whole-body imitation under physics constraints. The student receives only deployable observations, including joint states, base angular velocity, gravity direction, action

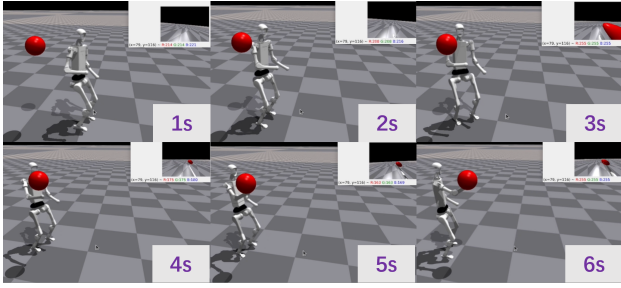


Fig. 5: Screenshots of reaching task under image input. The robot first searches for the ball when it is outside the field of view, then switches to reaching once the target becomes visible.

history, and the sparse targets generated by the planner. During runtime, the planner updates target trajectories at approximately 10 Hz, the student tracker computes joint targets at 30 Hz, and a PD controller outputs joint torques at 180 Hz. This asynchronous architecture separates long-horizon task planning from short-horizon stabilization and yields robust execution despite the planner’s lower update rate. It also avoids forcing the diffusion model to directly predict joint torques, which would greatly increase dimensionality and reduce portability.

IV. EXPERIMENTS

A. Setup

Experiments are conducted in Isaac Gym[18] with a Unitree H1 humanoid (19 DoF) and a head-mounted RGB camera. The task is to reach a target ball with the right hand. We evaluate two input settings: low-dimensional target positions and egocentric RGB images. The dataset contains 280 synthesized motion clips (50,410 frames).

The planner predicts future sparse head-and-hand trajectories in a receding-horizon manner. During deployment, the planner, student tracker, and PD controller run at approximately 10 Hz, 30 Hz, and 180 Hz, respectively. The observation, prediction, and execution horizons are set to $H_o = 4$, $H_p = 16$, and $H_a = 2 \sim 4$. The diffusion planner is trained with 100 denoising steps and uses 16 denoising steps at test time for faster inference.

B. Planner Success Rate and Qualitative Tracking Results

We evaluate the full hierarchical system on the ball-reaching task under both low-dimensional and image observations. Table I reports the success rate, execution time, and final reaching error on both seen and unseen targets.

We evaluate the full system on training and unseen targets using success rate, execution time, and final reaching error, with over 20 trials for each setting. A trial is counted as successful if the robot can locate the ball and complete the reaching behavior without getting stuck. Table I shows that, under low-dimensional position input, the system achieves a 100% success rate on both seen and unseen targets. Under image input, one round of DAgger improves performance to 88.89% on seen targets and 72.72% on unseen targets, while yielding lower final reaching error than the version without DAgger. These results indicate that the planner can generate

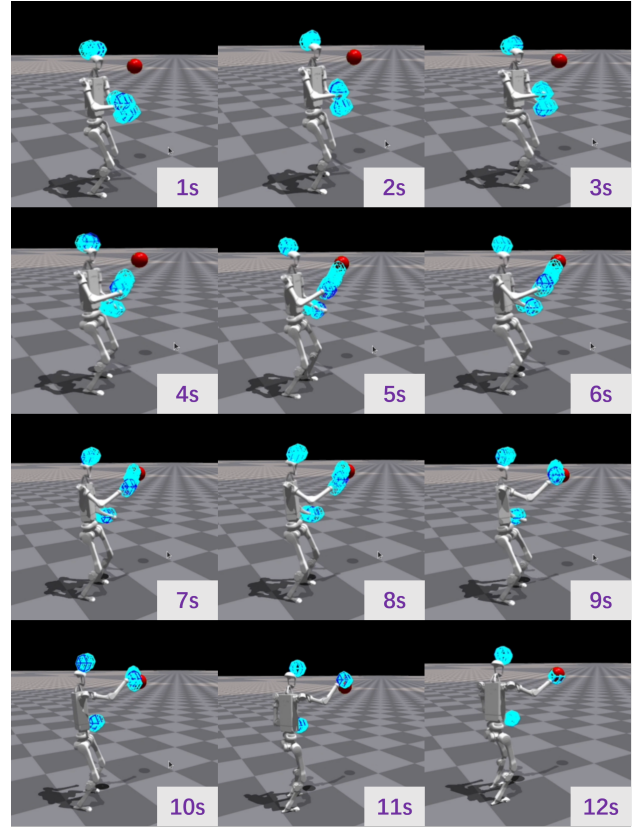


Fig. 6: Screenshots of the trajectory produced by diffusion policy. The planner continuously updates sparse keypoint trajectories, and the tracker executes them to guide the robot toward the ball.

effective goal-directed trajectories and that DAgger improves robustness.

As shown in Fig. 5, under image input, the robot first searches for the ball by turning and stepping backward when the object is outside the field of view, and then switches to goal-directed reaching once the ball appears in view. Fig. 6 further visualizes the sparse trajectory produced by the diffusion planner during this process: the light-blue markers denote the predicted trajectory sequence, while the dark-blue marker indicates the current target being tracked by the low-level controller.

C. Trajectory Tracking Results

We evaluate the student controller on two types of reference trajectories: dataset-aligned task trajectories and diffusion-generated trajectories under unseen task settings. Since the student controller serves as the low-level executor of the planner, its ability to track both types of references is critical for closed-loop deployment.

As shown in Table II, the student controller achieves a 100% success rate on both dataset-aligned and diffusion-generated trajectories, with average three-keypoint errors of 0.100 m and 0.125 m, respectively, indicating reliable tracking performance. Consistent with this, the last two rows of Table I show that removing guidance has little effect on success rate over unseen targets, but noticeably increases the final reaching error. This indicates that guidance mainly improves trajectory smoothness and temporal consistency,

TABLE I: Planner performance of the full hierarchical system.

Setting	Target Setting	Input Type	Avg. Init. Dist. (m)	Success Rate (%)	Avg. Time (s)	Final Error (m)
1-round DAGger	seen	Pos	0.8348	100.00	15.136	0.1257
1-round DAGger	unseen	Pos	1.0160	100.00	16.070	0.1698
w/o DAGger	seen	Img	0.8930	77.78	10.790	0.1630
w/o DAGger	unseen	Img	0.9770	60.00	13.860	0.1465
1-round DAGger	seen	Img	0.9520	88.89	11.921	0.0670
1-round DAGger	unseen	Img	1.0590	72.72	20.2233	0.1338
1-round DAGger w/o Guidance	unseen	Pos	1.0160	100.00	16.214	0.2315
1-round DAGger w/o Guidance	unseen	Img	1.0590	72.72	17.487	0.1846

TABLE II: Tracking performance of the student controller.

Setting	Avg. Length (frames)	Success Rate (%)	Avg. 3-Keypoint Error (m)
Dataset-traj.	165	100%	0.100
Diffusion-traj.	162	100%	0.125

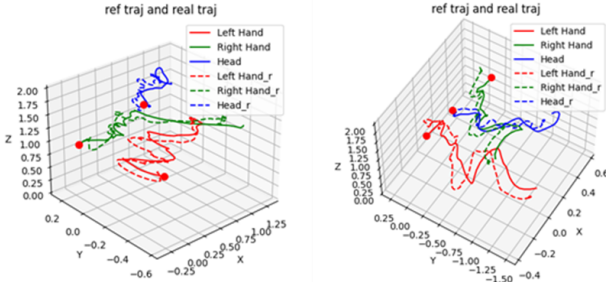


Fig. 7: Qualitative visualization of student-controller tracking for diffusion-generated trajectories. Solid lines denote reference trajectories, and dashed lines denote realized trajectories.

producing controller-friendly references that are easier to track accurately.

Fig. 7 provides a qualitative example of tracking a diffusion-generated trajectory. The realized trajectory remains close to the reference, showing that the student controller is sufficient for stable execution of diffusion-planned motions.

V. CONCLUSIONS

This paper presented a hierarchical framework for task-oriented humanoid reaching with scalable data generation, guided diffusion planning, and whole-body tracking control. By leveraging a physically grounded latent motion prior, goal relabeling, and OOD post-training, the proposed method improves both scalability and closed-loop robustness. Experiments on a Unitree H1 humanoid in Isaac Gym validate its effectiveness under both low-dimensional and image observations.

REFERENCES

- [1] L. Pan, Z. Yang, Z. Dou, W. Wang, B. Huang, B. Dai, T. Komura, and J. Wang, “Tokenhsi: Unified synthesis of physical human-scene interactions through task tokenization,” in *Proceedings of the Computer Vision and Pattern Recognition Conference*, 2025, pp. 5379–5391.
- [2] S. Shigemitsu, A. Goswami, and P. Vadakkepat, “Asimo and humanoid robot research at honda,” *Humanoid robotics: A reference*, vol. 55, p. 90, 2018.
- [3] Q. Sheng, Z. Zhou, J. Li, X. Mi, P. Xiang, Z. Chen, H. Xu, S. Jia, X. Wu, Y. Cui *et al.*, “A comprehensive review of humanoid robots,” *SmartBot*, vol. 1, no. 1, p. e12008, 2025.

- [4] L. Xiao, S. Lu, H. Pi, K. Fan, L. Pan, Y. Zhou, Z. Feng, X. Zhou, S. Peng, and J. Wang, “Motionstreamer: Streaming motion generation via diffusion-based autoregressive model in causal latent space,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, October 2025, pp. 10 086–10 096.
- [5] Z. Gu, Y. Chen, Z. Chai, A. Cueva, T. Nguyen, Y. Wu, H. Xue, M. Kim, I. Legene, F. Liu *et al.*, “Refine-dp: Diffusion policy fine-tuning for humanoid loco-manipulation via reinforcement learning,” *arXiv preprint arXiv:2603.13707*, 2026.
- [6] N. Mahmood, N. Ghorbani, N. F. Troje, G. Pons-Moll, and M. J. Black, “AMASS: Archive of motion capture as surface shapes,” in *International Conference on Computer Vision*, Oct. 2019, pp. 5442–5451.
- [7] Z. Luo, J. Cao, A. W. Winkler, K. Kitani, and W. Xu, “Perpetual humanoid control for real-time simulated avatars,” in *International Conference on Computer Vision (ICCV)*, 2023.
- [8] Z. Luo, J. Cao, J. Merel, A. Winkler, J. Huang, K. M. Kitani, and W. Xu, “Universal humanoid motion representations for physics-based control,” in *The Twelfth International Conference on Learning Representations*, 2024. [Online]. Available: <https://openreview.net/forum?id=OrOd8PxOO2>
- [9] G. Tevet, S. Raab, B. Gordon, Y. Shafir, D. Cohen-Or, and A. H. Bermano, “Human motion diffusion model,” *arXiv preprint arXiv:2209.14916*, 2022.
- [10] Y. Wu, K. Karunratanakul, Z. Luo, and S. Tang, “Uniphys: Unified planner and controller with diffusion for flexible physics-based character control,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025.
- [11] C. Chi, Z. Xu, S. Feng, E. Cousineau, Y. Du, B. Burchfiel, R. Tedrake, and S. Song, “Diffusion policy: Visuomotor policy learning via action diffusion,” *The International Journal of Robotics Research*, 2024.
- [12] G. Tevet, S. Raab, S. Cohan, D. Reda, Z. Luo, X. B. Peng, A. H. Bermano, and M. van de Panne, “CLOSD: Closing the loop between simulation and diffusion for multi-task character control,” in *The Thirteenth International Conference on Learning Representations*, 2025. [Online]. Available: <https://openreview.net/forum?id=pZISppZSTv>
- [13] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, “Proximal policy optimization algorithms,” *arXiv preprint arXiv:1707.06347*, 2017.
- [14] T. He, Z. Luo, X. He, W. Xiao, C. Zhang, W. Zhang, K. Kitani, C. Liu, and G. Shi, “Omnih2o: Universal and dexterous human-to-humanoid whole-body teleoperation and learning,” *arXiv preprint arXiv:2406.08858*, 2024.
- [15] M. Andrychowicz, F. Wolski, A. Ray, J. Schneider, R. Fong, P. Welinder, B. McGrew, J. Tobin, O. Pieter Abbeel, and W. Zaremba, “Hindsight experience replay,” *Advances in neural information processing systems*, vol. 30, 2017.
- [16] X. Liu, D. H. Park, S. Azadi, G. Zhang, A. Chopikyan, Y. Hu, H. Shi, A. Rohrbach, and T. Darrell, “More control for free! image synthesis with semantic diffusion guidance,” in *Proceedings of the IEEE/CVF winter conference on applications of computer vision*, 2023, pp. 289–299.
- [17] S. Ross, G. Gordon, and D. Bagnell, “A reduction of imitation learning and structured prediction to no-regret online learning,” in *Proceedings of the fourteenth international conference on artificial intelligence and statistics. JMLR Workshop and Conference Proceedings*, 2011, pp. 627–635.
- [18] V. Makoviychuk, L. Wawrzyniak, Y. Guo, M. Lu, K. Storey, M. Macklin, D. Hoeller, N. Rudin, A. Allshire, A. Handa, and G. State, “Isaac gym: High performance gpu-based physics simulation for robot learning,” 2021.